

MU XIE

OBJECTIVE

My research interests focus on autonomous systems, control, and learning-based optimization for robotics. Through my graduate and undergraduate studies, I have built a strong foundation in both theoretical analysis and practical system implementation, and I aim to advance the integration of learning and control for safe and efficient autonomous decision-making.

PERSONAL DETAILS

Phone	(215)605-8303
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Webpage	https://muxie2001.github.io/

EDUCATION

Master at University of Pennsylvania	2024.01 - 2025.05
GPA: 3.78/4.0	
<u>Relative Curriculum:</u> Learning in Robotics, Modern Convex Optimization, Advanced Robotics, F1Tenth Autonomous Racing	
Undergraduate at Zhejiang University (ZJU)	2019.09 - 2023.06
GPA: 3.56/4.0	
<u>Relative Curriculum:</u> Undergrad Math Foundation Courses, Analog Signal and System, Machine Learning, Numerical Analysis	
Undergraduate at University of Illinois, Urbana-Champaign (UIUC)	2019.09 - 2023.06
GPA: 3.26/4.0	
<u>Relative Curriculum:</u> Digital Signal Processing, Digital System, Electronic Circuits, Linear & Feedback Control System	

WORK EXPERIENCE

Internship at Fusion Array Co. Ltd	2023.07 - 2023.12
<i>Fusion Array Co. Ltd, Hangzhou, Zhejiang</i>	
<u>About:</u> Internship on FPGA development for ADC chip testing system.	
Research Assistant at X-Lab	2025.05 - Present
<i>University of Pennsylvania, Philadelphia</i>	
<u>About:</u> Research on autonomous system, safe control in robotics and learning to optimize.	

PUBLICATIONS

A Hybrid Learning-to-Optimize Framework for Mixed-Integer Quadratic Programming
V.-A. Le, M. Xie, and R. Mangharam, 2025. *Submitted to Learning for Dynamics and Control Conference.*

RESEARCHES & PROJECTS

Course Project: F1TENTH Autonomous Racing	2025.01 - 2025.05
<i>University of Pennsylvania, ESE Department, Philadelphia</i>	
<u>About:</u> Participated in three autonomous racing rounds using a full ROS 2 stack for perception, planning, and control. Developed a complete racing pipeline integrating multiple control and perception modules.	
<u>Responsible for:</u> Planning & Control: Implemented Wall-Follow and Gap-Follow algorithms with PID tracking, Pure Pursuit trajectory tracking, MPC-based optimal control, and dynamic obstacle avoidance.	

Perception: Designed a vehicle detection pipeline combining YOLO-based object detection with LiDAR clustering for obstacle localization.

Racing: Competed in two speed-trial events and one head-to-head race, achieving real-time control performance under tight hardware constraints.

Tools: ROS 2, Python/C++, RViz, TensorRT.

Research Project: Learning to Optimization for Mixed Integer Problem

2025.09 - Present

University of Pennsylvania, XLab, Philadelphia

About: Developed a differentiable optimization framework for solving mixed-integer quadratic programs (MIQP) in Model Predictive Control (MPC). Integrated neural network predictions with optimization layers for learning-to-optimize research.

Responsible for: Built a CVXPYLayer-based differentiable QP module enabling gradient flow through continuous relaxations; designed supervised and end-to-end learning pipelines to train neural networks predicting integer decision variables; benchmarked solver performance against Gurobi baselines and analyzed convergence under constraint relaxations.

Tools: PyTorch, CvxPy, Gurobi, NumPy

Research Project: Gaussian Process Dynamics with MPPI Control

2025.06 - 2025.09

University of Pennsylvania, XLab, Philadelphia

About: Investigated Gaussian Process (GP)-based vehicle dynamics modeling and its integration with Model Predictive Path Integral (MPPI) control for uncertainty-aware autonomous driving.

Responsible for: Trained GP regressors for nonlinear vehicle dynamics using multi-dimensional state-control datasets; integrated learned models into MPPI to perform uncertainty-weighted trajectory optimization; evaluated performance on simulated racing tracks with stochastic disturbances and compared to nominal MPC baselines.

Tools: PyTorch, GPyTorch, NumPy, ROS 2., JAX.

Course Project: LLM-Guided Navigation for F1TENTH

2024.09 - 2024.12

University of Pennsylvania, ESE Department, Philadelphia

About: Explored the use of Large Language Models (LLMs) for high-level navigation and control in autonomous racing tasks, connecting natural-language commands to low-level driving primitives.

Responsible for: Designed a Python interface for translating natural-language goals into control policies using LLM reasoning; implemented task-driven decision-making combining symbolic reasoning and traditional control; tested the pipeline on simulated and real-world F1TENTH environments for interactive control and scenario planning.

Course Project: Quadrotor Control Paper Reproduction

2025.01 - 2025.05

University of Pennsylvania, ESE Department, Philadelphia

About: Reproduced and validated results from a published nonlinear quadrotor control paper, implementing full-state trajectory tracking and stability analysis in simulation.

Responsible for: Implemented nonlinear attitude and position controllers using cascaded feedback design; implemented algorithms for grid-world path planning and sensor fusion; validated trajectory-tracking accuracy and robustness under parametric model uncertainties; conducted numerical experiments to verify control stability and performance metrics.

Course Project: Image-to-GPS Regression

2024.10 - 2024.12

University of Pennsylvania, ESE Department, Philadelphia

About: Implemented a vision-transformer to estimate geographic coordinates from street imagery, investigating the feasibility of direct image-to-location regression.

Responsible for: Designed and trained vision-transformer models for continuous geolocation prediction using street image datasets; collected street image datasets for different urban environments; compared regression accuracy to baseline res-net methods.

Tools: PyTorch, OpenCV, NumPy.

SKILLS

Programming Languages	C/C++, PYTHON, TEX, MATLAB, VERILOG HDL
Development Tool	ROS 2, PYTORCH, JAX, QUARTUS, CADENCE SPB